



Car Rental Platform

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1. Introduction & System Overview

1.1 Project Introduction

The Car Rental Platform is a full-stack web application developed using **ReactJS, NodeJS + Express, and MySQL**.

The purpose of the system is to allow users to:

- Register and log in securely
- Browse available cars
- View detailed car information
- Book cars for selected dates
- View their booking history

The system follows a **three-tier architecture**:

React Frontend → Express API → MySQL Database

1.2 System Architecture Overview

The application is structured as follows:

- **Frontend:** ReactJS with routing and components

- **Backend:** NodeJS + Express REST API
 - **Database:** MySQL relational database
 - **Authentication:** JWT-based authentication
 - **Validation:** Client-side and server-side validation
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2. Front-End (ReactJS) Explanation

2.1 React Pages

Our system includes the following pages:

1. Home
2. Login
3. Register
4. Dashboard
5. Cars Catalogue
6. Car Details
7. Booking Page
8. My Bookings
9. Admin Panel

- ✓ Routing implemented using React Router
 - ✓ State management using `useState` and `useEffect`
 - ✓ Axios used for API communication
 - ✓ Client-side validation implemented
-

2.2 Example UI Screens

Home Page

Quick Rent

Home

Cars

My Bookings

London

9°C

moh@gmail.com

Logout

Unlock Your Adventure

Choose your car, pick your dates, and drive your dreams.

Pick Up

Return

Car Type

Max €

dd/mm/yyyy

dd/mm/yyyy

Any

€



Rent by Brand


Toyota


Ford


Tesla


Volkswagen


BMW


Mercedes


Hyundai


Audi

Rent by Body Type


SUV


Wagon


Sedan


Convertible


Coupe

Most Searched Vehicles



Ford focus

€40.00 / day

5

Auto

Petrol

View Details



Ford Fiesta

€40.00 / day

5

Manual

Petrol

View Details



Hyundai i20

€42.00 / day

5

Manual

Petrol

View Details

Cars Catalogue Page

 Quick Rent

[Home](#)

[Cars](#)

[My Bookings](#)

London

9°C

moh@gmail.com

Logout

Cars

25 results

Any Brand

Any Model

Any Type

Any Transmission

Any Fuel

Max €

Search

Reset



Ford Fiesta

€40.00 / day

5

ManualPetrol

View Details



Ford focus

€40.00 / day

5

AutoPetrol

View Details



Hyundai i20

€42.00 / day

5

ManualPetrol

View Details



Toyota Yaris

€45.00 / day

5

AutoPetrol

View Details



Renault Capture

€50.00 / day

5

AutoPetrol

View Details



Toyota Corolla

€55.00 / day

5

AutoHybrid

View Details



Ford Focus

€56.00 / day

5

ManualDiesel

View Details



Volkswagen Golf

€60.00 / day

5

ManualDiesel

View Details



Audi A3

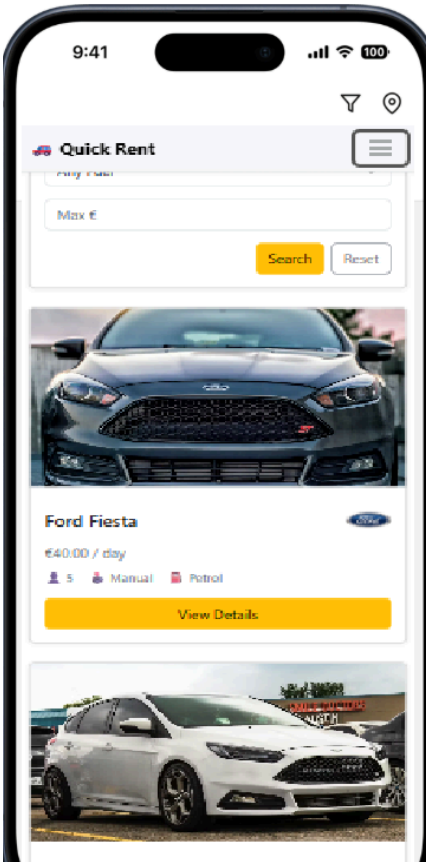
€65.00 / day

5

AutoPetrol

View Details

Mobile UI



Quick Rent

HomeCarsMy Bookings

London9°C

moh@gmail.comLogout



Mercedes AMG GT

CoupeAutomaticPetrol

2 seatsAutomaticPetrol

€150.00 / day

Book Now

Back to Cars

Why choose this car?

The Mercedes AMG GT is ideal for both city driving and longer journeys. It offers a comfortable interior, smooth automatic transmission, and reliable petrol performance — making it a great rental choice for any trip.

Included in your rental

✓ Unlimited mileage

✓ Free cancellation

✓ Basic insurance included

✓ 24/7 roadside assistance

Rental information

Minimum driver age: 21 years

Valid driving license required

Security deposit may apply

Quick Rent

HomeCarsMy Bookings

London9°C

Mmoh@gmail.comLogout

Booking Details

Start Date (DD/MM/YY)20/02/26

End Date (DD/MM/YY)DD/MM/YY

Emailmoh@gmail.com

February 2026

12345678910111213141516171819202122232425262728

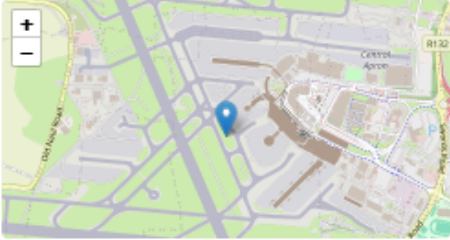
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Summary

Days0

Total€0

Pick-up Location (Dublin Airport)



Lat: 53.42640 | Lng: -6.24990

Quick Rent Home Cars My Bookings London 9°C moh@gmail.com Logout

Complete Payment

Card

Secure, fast checkout with Link

Card number: 1234 1234 1234 1234

Expiration date: MM / YY

Security code: CVC

Country: Ireland

Pay Now

Features:

- Date selection
- Validation of dates
- Booking confirmation#
- Payment using Stripe Sandbox

Amin Dashboard UI

Quick Rent Admin Dashboard Dublin 8°C med@gmail.com Logout

Admin Dashboard

Admin Panel

- Dashboard
- Bookings
- Cars
- Users

Logout

Users: 8

Cars: 25

Active Bookings: 3

Revenue: €444.00

Cancelled: 1

2.3 React Technical Implementation

- Components structured inside `/components`
- Pages inside `/pages`
- API calls inside `backend/server.js`
- Protected routes implemented for authenticated users
- JWT stored securely in `localStorage`
- `useEffect` used to fetch cars and bookings

3. Back-End (NodeJS + Express) Explanation

3.1 REST API Endpoints

The backend was developed as a separate NodeJS project using Express.

Authentication Routes:

- POST `/api/auth/register`
- POST `/api/auth/login`

Car Routes:

- GET `/api/cars`
- GET `/api/cars/:id`

Booking Routes:

- POST `/api/bookings`
- GET `/api/bookings/user/:id`

3.2 Server-Side Validation

Implemented using:

- Required field checks
- Email format validation
- Password length validation
- Date conflict checking for bookings

Example:

- Prevents double booking of same car for overlapping dates.

3.3 Authentication

- JWT token generated upon login
 - Middleware verifies token
 - Protected routes require valid token
 - Passwords hashed using bcrypt
-

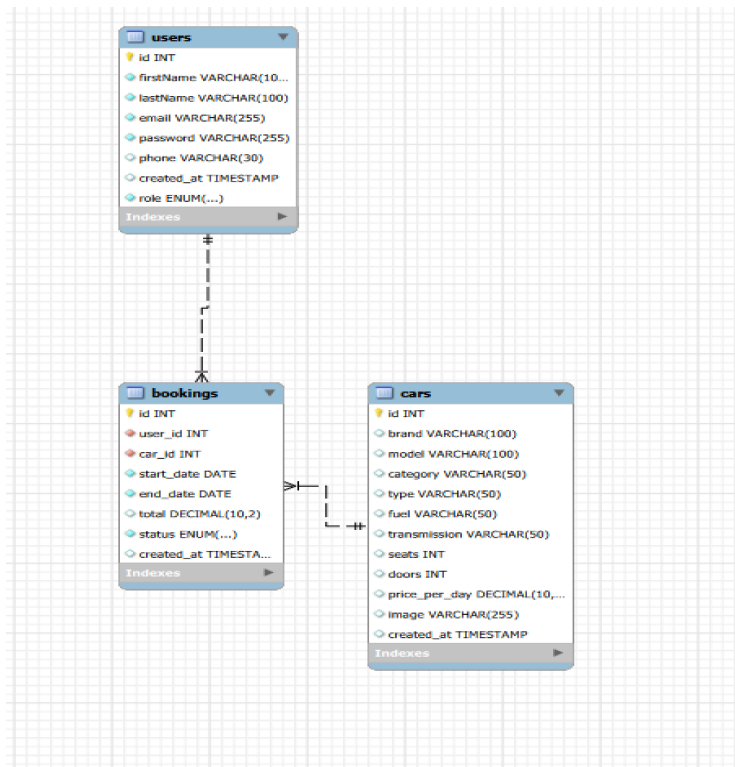
4. Database Design (MySQL)

4.1 Entities (Minimum 4 Required ✓)

The system contains the following entities:

1. Users
 2. Cars
 3. Bookings
 4. Payments (if implemented)
 - 5.
-

4.2 ERD Diagram



4.3 Relationships

- One User → Many Bookings
- One Car → Many Bookings
- Booking contains:
 - user_id (FK)
 - car_id (FK)
 - start_date
 - end_date
 - total_price

Primary Keys:

- user_id
- car_id
- booking_id

Foreign Keys:

- booking.user_id → users.user_id
 - booking.car_id → cars.car_id
-

4.4 CRUD Operations Implemented

- ✓ Create booking
 - ✓ Read cars
 - ✓ Read bookings
 - ✓ Register user
 - ✓ Update (if admin added)
 - ✓ Delete booking (if implemented)
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5. System Integration

The system connects as follows:

1. React sends request via Axios
2. Express API receives request
3. Express queries MySQL
4. MySQL returns data
5. Express sends JSON response
6. React updates UI state

This demonstrates full-stack integration.

6. Challenges Faced

- Handling date overlap logic
 - Protecting private routes
 - Managing state after booking
 - Connecting MySQL properly
 - Debugging CORS issues
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7. Conclusion

The Car Rental Platform successfully meets all module requirements :

- ✓ React Frontend (8+ pages)
- ✓ NodeJS + Express REST API
- ✓ MySQL with 4+ entities
- ✓ Authentication (JWT)
- ✓ Client & Server validation
- ✓ CRUD functionality

The project demonstrates understanding of full-stack web development and API integration.



Video Demo Structure (6 minutes)

Youtube link: <https://youtu.be/hNcY0FQHxfA>
